

RAJALAKSHMI ENGINEERING COLLEGE

An Autonomous Institution, Affiliated to Anna University

Rajalakshmi Nagar, Thandalam – 602 105

Web Technology And Mobile Application

Experiment 2

2026-2027

By

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AIM:

To write JavaScript code to validate user input in a registration form. The validation will ensure that the data entered by the user meets specific criteria for each field. This will help in maintaining data integrity and prevent invalid submissions.

ALGORITHM:

1. **Start**
2. Get the values of all fields from the registration form:
 - First Name
 - Password
 - Email
 - Mobile Number
 - Address
3. **Validate First Name**
 - Check if it contains only alphabets.
 - Check if the length is **≥ 6 characters**.
 - If not valid, show an error message and stop.
4. **Validate Password**
 - Check if the length is **≥ 6 characters**.
 - If not valid, show an error message and stop.
5. **Validate Email**
 - Check if it matches the standard pattern **name@domain.com**
 - If not valid, show an error message and stop.
6. **Validate Mobile Number**
 - Check if it contains only digits.
 - Check if the length is exactly **10 digits**.
 - If not valid, show an error message and stop.
7. **Validate Last Name**
 - Check if it is not empty.
 - If empty, show an error message and stop.
8. **Validate Address**
 - Check if it is not empty.
 - If empty, show an error message and stop.
9. If all fields are valid:
 - Display message: **“Form Submitted Successfully!”**
10. **End**

CODE:

HTML code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>ORRS Registration</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link rel="icon" type="image/webp" href="logo.webp">
  <link rel="stylesheet" href="styles.css">
</head>
<body>

<div class="container">
  <h2>Sign Up Here</h2>
  <form id="registrationForm" novalidate>
    <div class="form-group">
      <label for="firstName">First Name:</label>
      <input type="text" id="firstName" placeholder="First Name">
      <span class="error" id="firstNameError"></span>
    </div>
    <div class="form-group">
      <label for="email">Email:</label>
      <input type="email" id="email" placeholder="E-mail">
      <span class="error" id="emailError"></span>
    </div>
    <div class="form-group">
      <label for="password">Password:</label>
      <input type="password" id="password" placeholder="Password">
      <span class="error" id="passwordError"></span>
    </div>
    <div class="form-group">
      <label for="mobile">Mobile Number:</label>
      <input type="tel" id="mobile" placeholder="10-digit Mobile Number">
      <span class="error" id="mobileError"></span>
    </div>
    <div class="form-group">
      <label for="address">Address:</label>
      <textarea id="address" rows="4" placeholder="Address"></textarea>
      <span class="error" id="addressError"></span>
    </div>
    <button type="submit" class="submit-btn">Register</button>
  </form>
</div>
<script src="script.js"></script>
<script src="https://cdn.jsdelivr.net/npm/sweetalert2@11"></script>
</body>
</html>
```

CSS code:

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  background-image: url('background\ orrs.avif');
  background-size: cover;
  background-position: center;
  background-repeat: no-repeat;
  background-attachment: fixed;
  background-color: whitesmoke;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  padding: 20px;
}

.container {
  background: rgba(0, 0, 0, 0.4);
  padding: 40px;
  border-radius: 30px;
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.2);
  width: 100%;
  max-width: 500px;
  transition: background 0.4s ease;
}

.container:hover {
  background: rgba(0, 0, 0, 0.8);
}

h2 {
  text-align: center;
  color: whitesmoke;
  margin-bottom: 30px;
  font-size: 28px;
}

.form-group {
  margin-bottom: 10px;
}

label {
  display: block;
  margin-bottom: 8px;
  color: whitesmoke;
  font-weight: 600;
  font-size: 14px;
}

input[type="text"],
input[type="email"],
input[type="password"],
input[type="tel"],
textarea {
  width: 100%;
  padding: 12px 15px;
  border: 2px solid #ddd;
  border-radius: 10px;
  font-size: 14px;
  transition: border-color 0.3s;
  outline: none;
}
```

```
input[type="text"]:focus,
input[type="email"]:focus,
input[type="password"]:focus,
input[type="tel"]:focus,
textarea:focus{
    border: 1px solid;
    border-image: linear-gradient(to right, red, blue) 1;
    border-radius: 3px;
}
input.invalid,
textarea.invalid {
    border-color: #e74c3c;
}
input.valid,
textarea.valid {
    border-color: #2ecc71;
}
textarea {
    resize: vertical;
    font-family: Arial, sans-serif;
}
.error {
    display: block;
    color: #e74c3c;
    font-size: 12px;
    margin-top: 5px;
    min-height: 18px;
}
.submit-btn {
    width: 100%;
    padding: 14px;
    background: lightgreen;
    color: black;
    border: none;
    border-radius: 5px;
    font-size: 16px;
    font-weight: 600;
    cursor: pointer;
    transition: transform 0.2s, box-shadow 0.2s;
}
.submit-btn:hover {
    transform: translateY(-2px);
    box-shadow: 0 5px 15px rgba(102, 126, 234, 0.4);
}
.submit-btn:active {
    transform: translateY(0);
}
@media (max-width: 600px) {
    .container {
        padding: 30px 20px;
    }
    h2 {
        font-size: 24px;
    }
}
```

JAVASCRIPT code:

```
document.getElementById('registrationForm').addEventListener('submit', function (e) {
    e.preventDefault();
    clearErrors();

    const firstName = document.getElementById('firstName').value.trim();
    const email = document.getElementById('email').value.trim();
    const password = document.getElementById('password').value;
    const mobile = document.getElementById('mobile').value.trim();
    const address = document.getElementById('address').value.trim();
    let isValid = true;
    if (!validateFirstName(firstName)) isValid = false;
    if (!validateEmail(email)) isValid = false;
    if (!validatePassword(password)) isValid = false;
    if (!validateMobile(mobile)) isValid = false;
    if (!validateAddress(address)) isValid = false;
    if (isValid) {
        Swal.fire({
            title: 'Success!',
            text: 'Registration Successful!',
            icon: 'success',
            confirmButtonText: 'OK'
        });
        document.getElementById('registrationForm').reset();
        clearErrors();
    }
});

function validateFirstName(firstName) {
    const field = document.getElementById('firstName');
    const error = document.getElementById('firstNameError');
    const pattern = /^[A-Za-z]{2,}$/;
    if (!pattern.test(firstName)) {
        error.textContent = "First name must contain only letters (min 2 characters)";
        setInvalid(field);
        return false;
    }
    setValid(field);
    error.textContent = "";
    return true;
}

function validateEmail(email) {
    const field = document.getElementById('email');
    const error = document.getElementById('emailError');
    const pattern = /^[^\s@]+@[^\s@]+\.[^\s@]{2,}$/;
    if (!pattern.test(email)) {
        error.textContent = "Enter a valid email (example@domain.com)";
        setInvalid(field);
        return false;
    }
    setValid(field);
    error.textContent = "";
    return true;
}

function validatePassword(password) {
    const field = document.getElementById('password');
    const error = document.getElementById('passwordError');
    if (password.length < 6) {
        error.textContent = "Password must be at least 6 characters";
        setInvalid(field);
        return false;
    }
}
```

```

    setValid(field);
    error.textContent = "";
    return true;
}

function validateMobile(mobile) {
    const field = document.getElementById('mobile');
    const error = document.getElementById('mobileError');
    const pattern = /^[0-9]{10}$/;
    if (!pattern.test(mobile)) {
        error.textContent = "Mobile number must contain exactly 10 digits";
        setInvalid(field);
        return false;
    }
    setValid(field);
    error.textContent = "";
    return true;
}

function validateAddress(address) {
    const field = document.getElementById('address');
    const error = document.getElementById('addressError');
    if (address.length < 5) {
        error.textContent = "Address must be at least 5 characters";
        setInvalid(field);
        return false;
    }
    setValid(field);
    error.textContent = "";
    return true;
}

function setInvalid(field) {
    field.classList.add("invalid");
    field.classList.remove("valid");
}

function setValid(field) {
    field.classList.add("valid");
    field.classList.remove("invalid");
}

function clearErrors() {
    document.querySelectorAll(".error").forEach(e => e.textContent = "");
    document.querySelectorAll("input, textarea").forEach(input => {
        input.classList.remove("valid", "invalid");
    });
}

document.getElementById('firstName').addEventListener('blur', function () {
    validateFirstName(this.value.trim());
});

document.getElementById('email').addEventListener('blur', function () {
    validateEmail(this.value.trim());
});

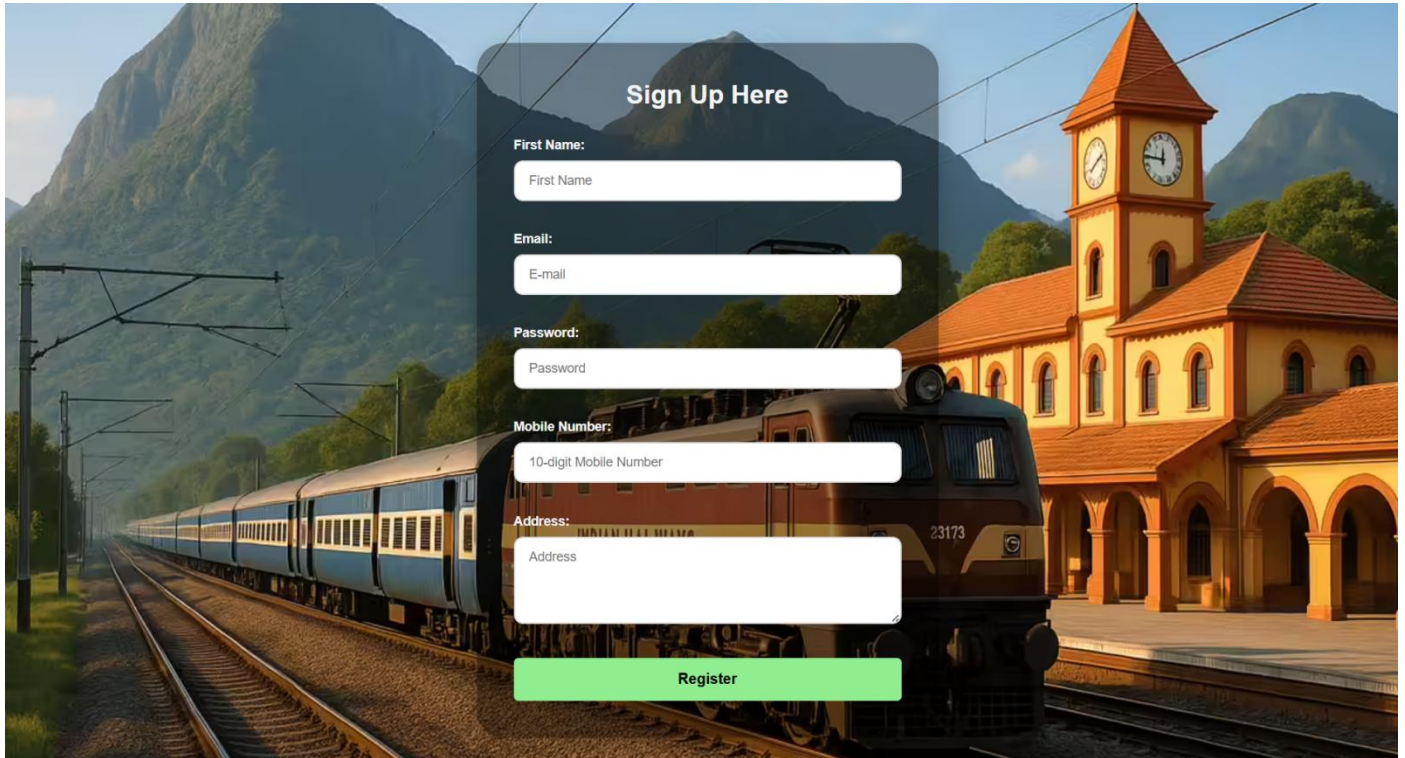
document.getElementById('password').addEventListener('blur', function () {
    validatePassword(this.value);
});

document.getElementById('mobile').addEventListener('blur', function () {
    validateMobile(this.value.trim());
});

document.getElementById('address').addEventListener('blur', function () {
    validateAddress(this.value.trim());
});

```

OUTPUT:



Sign Up Here

First Name:

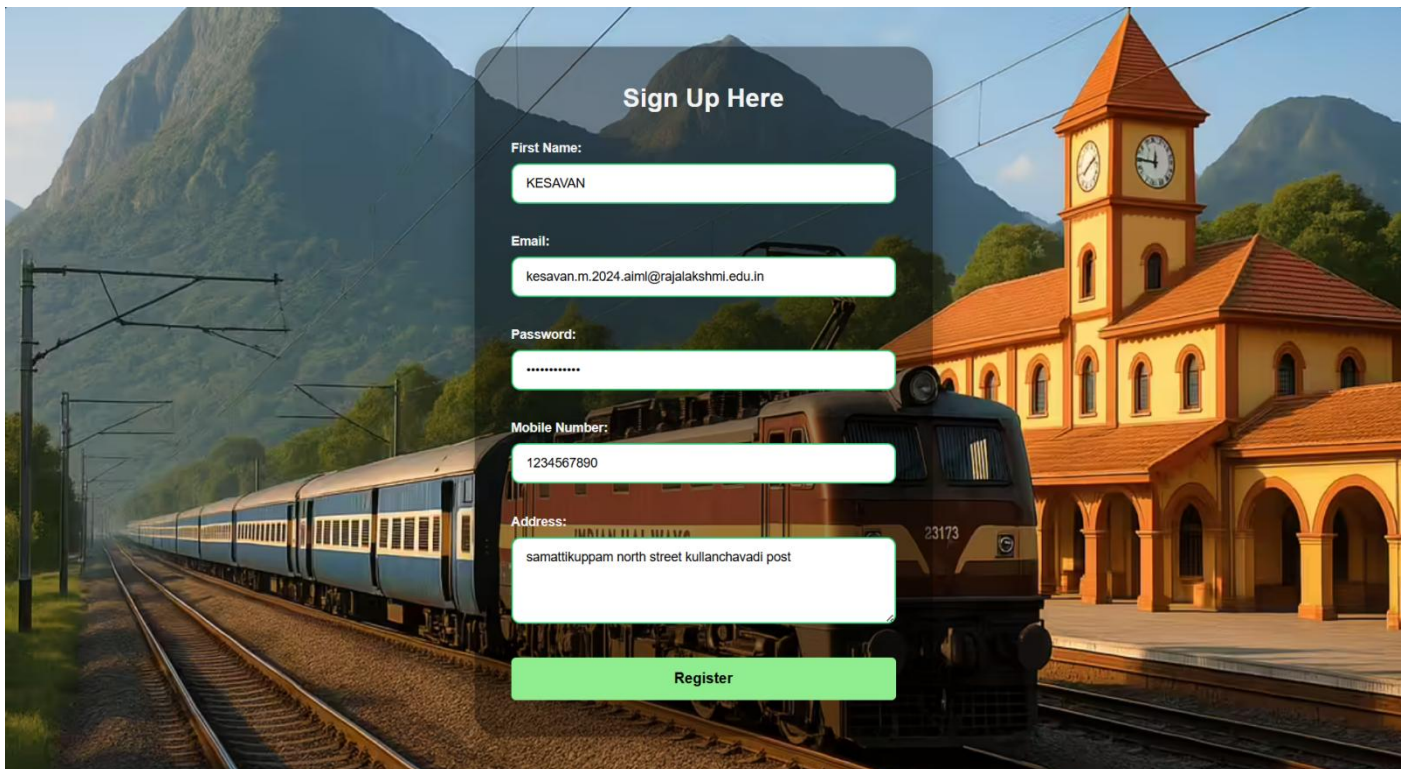
Email:

Password:

Mobile Number:

Address:

[Register](#)



Sign Up Here

First Name:

Email:

Password:

Mobile Number:

Address:

[Register](#)

RESULT:

Html Page is created with embedded Chennai map along with hot spot, frames & links.